

Jiali Shan

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Education

- Ph.D. Fudan University**, School of Mathematical Sciences Shanghai, China
 • Research areas: randomized numerical linear algebra, total least squares, backward error analysis, large-scale scientific computing. 2025 – present
- B.S. Fudan University**, Mathematics and Applied Mathematics Shanghai, China
2021 – 2025

Experience

- University of California, Berkeley**, Exchange Student, Department of Mathematics Berkeley, CA, USA
 • Coursework: all courses graded A+/A. Aug 2024 – Dec 2024

Publications

- Efficient estimate for the optimal backward error of the multidimensional total least squares** 2026
 Jiali Shan, Yimin Wei
[10.1137/25M1788312](#) (SIAM Journal on Matrix Analysis and Applications, 47(2), 467–672)
- Optimal backward error of a total least squares and its randomized algorithms** 2025
 Jiali Shan, Yimin Wei
[10.1137/25M1749657](#) (SIAM Journal on Matrix Analysis and Applications, 46(3), 2116–2139)
- Neural networks for total least squares solution of the time-varying linear systems** 2025
 Xuezhong Wang, Jiali Shan, Yimin Wei
[10.1007/s40314-024-03070-1](#) (Computational and Applied Mathematics, 44(3), 106)

Funding

- National Natural Science Foundation of China (NSFC)**
 • Role: Project leader (Youth Student Basic Research Project)
 • Grant: 123B1008
 • duration: 2023-2025
 • topic: Randomized Algorithms for Toeplitz Total Least Squares Problems

Awards

- 2025 ICCM Undergraduate Paper Award** Jan 2026
- 2026 ICBS Undergraduate Paper Award** Aug 2026

Skills

- Programming:** MATLAB, Python, C, LaTeX
- Languages:** Mandarin Chinese (native), English (fluent), Japanese (basic)

Talks

- Backward analysis of the total least squares and its Randomized Algorithms**
 • The 8th Peking University Symposium for Outstanding PhD Students in Computational and Applied Mathematics, Peking University, Nov. 2025.
 • The 7th Luojia Doctoral Forum in Mathematics, Wuhan University, Apr. 2026.
 • MoRat summer school, EPFL & ETH, Sep. 2026.